Project 32k-XAI: Master Technical Specification

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Critical Revision Note: This document consolidates all architectural, psychological, and computational specifications into a single self-contained reference. It incorporates mandatory corrections identified during the DARPA-grade adversarial audit and Sprint 0 Verification & Recovery protocol. It resolves all contradictions between the previous 32k-XAI specification and the published DT³ article (Khorshidi & Mansouri Jamshidi, 2026, SSRN 7306299). No progression beyond Sprint 0 is permitted until Gates G1–G9 are documented and passed.

Part I: Project Overview and Scientific Rationale

1.1 Project Identity

Project 32k-XAI is a research platform for simulating a synthetic population of 32,000 autonomous cognitive agents, each governed by a transparent, mechanistically interpretable decision-making architecture parameterized by empirically grounded psychological variables. The platform's purpose is to enable causal, glass-box interrogation of emergent social phenomena—trust breakdown, economic segregation, cooperation collapse—by tracing macro-level outcomes to specific cognitive parameters at the individual level.

Core epistemological position: The platform does **not** claim to be “100% transparent” in an absolute sense, nor “strictly deterministic” as a population. It is **within-agent deterministic, between-agent stochastic**, parameterized by a fixed random seed. Transparency is **uncertainty-aware**: every mechanistic trace carries documented measurement uncertainty, reliability flags, and provenance annotations.

1.2 The Problem with Contemporary MARL

The contemporary landscape of Multi-Agent Reinforcement Learning (MARL) is fundamentally flawed for sociological and psychological simulation. Deep Neural Networks (DNNs) act as impenetrable black boxes, representing human behavior as homogeneous, latent-space approximations devoid of empirical psychological grounding. No action can be

causally traced to a specific cognitive mechanism. Emergent social phenomena—betrayal, trust, segregation—are statistical artifacts of gradient descent, not consequences of theoretically specified psychological parameters.

1.3 The 32k-XAI Paradigm Shift

Project 32k-XAI proposes a scientifically constrained alternative: a **mechanistically interpretable multi-agent platform** in which every agent's "brain" is a unified system of transparent mathematical functions—Temporal Difference learning, forward-planning MDP solvers, Fehr-Schmidt social utility, Softmax action selection—mapped directly to a psychometric parameter vector. There are no neural networks in the decision-making loop. Every action, betrayal, trust allocation, and system failure can be traced to a specific cognitive parameter, conditional on that parameter's documented measurement uncertainty.

1.4 Key Design Principles

Principle	**Specification**
No black boxes	DNNs and LLMs are strictly banned from the decision-making loop. Every decision is a closed-form or iterative mathematical function.
Empirical grounding	Population parameters are tethered to real human covariance structures via a Vine Copula, not arbitrary synthetic generation.
Uncertainty-aware transparency	Every XAI trace includes confidence intervals, recovery metrics, and reliability flags. No bare point estimates are reported as ground truth.
Recovery-gated scale-up	No 32k simulation is run until all parameters pass rank-order recovery $\rho > 0.7$.
Reproducibility	Every agent has a unique ID and seed. Trial-level data is stored. Generator version and seed list are immutable.
DT³ alignment	The Dark Triad is modeled as a bifactor structure (general Dark Core + specific factors), consistent with the published DT³ article.

Part II: Scientific Foundations

2.1 The Dark Triad as a Multi-Layered Structure

The published DT³ article (Khorshidi & Mansouri Jamshidi, 2026) triangulated across five computational layers to test the distinctiveness and convergence of Machiavellianism, psychopathy, and narcissism. The key findings that directly constrain the 32k-XAI architecture are:

Finding 1: Bifactor structure, not three independent dimensions. Confirmatory factor analyses across multiple samples consistently show that the best-fitting model for both the SD3 and DTDD is a bifactor model with a general Dark Core factor plus specific factors. A meta-analytic SEM study of 11 SD3 studies and 5 DD studies in the United States and Canada found that conventional correlated-factor and orthogonal bifactor models fail to explain the irregular correlations among items; for Dirty Dozen items, a bifactor-(S·I-1) model is more suitable, and the complex structure of the SD3 necessitates item revision to enhance reliability and discriminant validity . The DT³ article's own CFA found three factors (CFI = 0.941, RMSEA = 0.077) but with high latent correlations (r = 0.62, 0.61).

Finding 2: Dark Core Collapse. When the Dark Triad items are embedded in the full 88-item personality space (Big Five + empathy + Dark Triad), all Dark Triad items merge into a single Louvain community. The boundaries of the three traits are observable only when artificially isolated from broader personality context; once embedded in a wider nomological space, these boundaries dissolve into a unidimensional, shared structure.

Finding 3: Narcissism Fracture. Hierarchical clustering on item-text embeddings yielded ARI = 0.5045 relative to theoretical labels. Psychopathy and Machiavellianism items each achieved 100% cluster purity; only 75% of narcissism items fell within their dedicated cluster, with the remaining item joining the Machiavellianism cluster. Machiavellianism and narcissism converge on agreeableness as their shared behavioral tipping point, while psychopathy depends specifically on empathy—consistent with a two-branch structure: an “instrumental disagreeableness” pole (Machiavellianism, narcissism) and an “affective callousness” pole (psychopathy).

Finding 4: Limited explanatory power. Out-of-sample R^2 ranged between 0.13 and 0.23, indicating that supervised models explain only a limited portion of trait variance. Psychopathy showed consistently lower reliability (ω = 0.638 in the student sample; ICC(2,1) = 0.637 for test-retest), compared to Machiavellianism (ICC = 0.826) and narcissism (ICC = 0.836).

Architectural implication: The Dark Triad must be modeled as a bifactor structure: one general Dark Core dimension (*D*) plus three specific factors (*S_M*, *S_P*, *S_N*), not as three independent Copula dimensions. The Copula must be estimated on the full item space (or its psychometric equivalent), not on the isolated 12-item DTDD subscale.

2.2 DASS-21: Bifactor Structure, Not Three Orthogonal Factors

The DASS-21 was originally designed with three correlated factors (depression, anxiety, stress). However, confirmatory factor analyses across eight countries (Brazil, Canada, Hong Kong, Romania, Taiwan, Turkey, UAE, USA; N = 2,580) compared unidimensional, three-correlated-factors, higher-order, and bifactor models. The bifactor model—with three specific factors (depression, anxiety, stress) and one general factor (general distress)—presented the best fit within each country . Results suggested the DASS-21 can be used as a unidimensional scale, and the authors concluded it would best be used as a general score of distress rather than three separate factors . Bifactor modeling in traumatic brain injury samples revealed a dominant general distress factor accounting for 89% of the systematic variance in DASS-21 total scores .

Architectural implication: Dynamic modulation of the arbitration weight *w_MB* under stress must use the **general distress factor** from the bifactor model, not three orthogonal factors. The specific factors (depression, anxiety, stress) may be retained as secondary modulators with documented lower reliability.

2.3 Two-Step Task: Daw vs. Kool Paradigm

The Daw two-step task is the canonical paradigm for estimating model-based vs. model-free arbitration (*w_MB*). However, the evidence on parameter recovery is not clean. For the Daw task, model comparison shows that AIC favors the hybrid model but BIC favors the pure model-free model. The modified Kool paradigm, in which model-free agents show interaction effects qualitatively similar to model-based agents, allows reinforcement learning models to distinguish model-based and model-free behavior without further modification . Kool et al. suggest fitting reinforcement learning models to distinguish subtle differences between model-free and model-based behavior in such cases .

Architectural implication: The Kool two-step paradigm is adopted as the primary source of *w_MB* values, with the Daw paradigm retained for comparison and validation. Parameter recovery must demonstrate $\rho > 0.6$ for *w_MB* in at least one paradigm before 32k scale-up.

2.4 Fehr-Schmidt Identifiability

The Fehr-Schmidt inequity aversion model uses two parameters: α (envy, disadvantageous inequity aversion) and β (guilt, advantageous inequity aversion). However, the parameters are under-identified and may float to a considerable degree . Fehr and Schmidt themselves altered the parameter distribution across subsequent papers, using a 40–60 distribution in later analyses that reclassified mildly inequity-averse subjects as money-maximizers . This distributional difference is meaningful: subjects with mild inequity aversion are not behaviorally identical to pure money-maximizers .

Architectural implication: If recovery testing shows that α_i and β_i cannot be reliably recovered from the available Ultimatum/Trust trial counts, the utility function is simplified to a single-parameter inequity aversion index, and this simplification is flagged in every XAI trace.

2.5 Stop-Signal and Drift-Diffusion Identifiability

The Drift-Diffusion Model (DDM) decomposes decision-making into drift rate (v), boundary separation (a), and non-decision time (t_0). DDM parameters are identifiable only when trial-level response time (RT) distributions are available. If only accuracy/SSRT data are available, DDM parameters are not identifiable, and the Stop-Signal Reaction Time (SSRT) index must be used instead. SSRT is computed as the mean of the stop-signal delay plus the mean go-trial RT minus the mean stop-trial RT, using the integration method .

Architectural implication: Audit `XAI_Risk_Impulsivity_Data` for trial-level RTs before attempting DDM fitting. If RTs are missing, use SSRT as the inhibitory control index.

2.6 Hierarchical Bayesian Parameter Recovery

Hierarchical Bayesian Analysis (HBA) provides simultaneous estimation of individual- and group-level parameters. However, hierarchical modeling does not guarantee unbiased parameter estimation. The choice of priors and hyperparameters can bear significant impact on parameter estimation and reliability, and the hierarchical nature of the estimation procedure can distort parameters toward high-precision individuals. Katahira et al. (2024), cited in Sabihe's v1.2 specification, warns that hierarchical modeling can reduce individual-level parameter reliability when true between-subject variance is large.

Recovery protocol: Generate 500–1,000 synthetic subjects per pilot condition with known parameters. Simulate trial-level data. Fit exactly the model family intended for the human/agent data. Report bias, RMSE, rank-order recovery (Spearman ρ), posterior coverage, and parameter correlations. Reject any parameter with rank-order recovery $\rho < 0.7$ before 32k scale-up.

2.7 Vine Copula for Non-Gaussian Dependence

The Gaussian Copula assumes that after marginal transformation, the dependence structure is fully captured by a single correlation matrix Σ on latent normal variables. This is inadequate for psychological data, which exhibit non-Gaussian tail dependence, skewness beyond what marginal transformations can fix, and heteroscedasticity. Standard multivariate copulas (Gaussian, Student-t) lack flexibility for high-dimensional systems ($d > 3$), imposing rigid, symmetric dependence structures . Vine copulas decompose the joint distribution into a cascade of bivariate pair-copulas, allowing non-Gaussian tail dependence and asymmetric interactions . The VITA (VIne To Anything) method generalizes the NORTA (NORmal To Anything) method for non-normal data with user-specified marginals and covariance .

Architectural implication: The generative engine is upgraded from Gaussian Copula to Vine Copula (D-Vine or R-Vine). Pair-copula families (Gaussian, Student-t, Clayton, Gumbel, Frank) are selected per edge via BIC/AIC. Clayton copulas capture lower-tail dependence; Gumbel copulas capture upper-tail dependence.

Part III: Parameter Dictionary (21 Parameters)

All parameters are defined with range, recommended transform, population sampling method, and evidence source. No parameter may be assigned a value outside its specified range. No parameter may be interpreted as a clinical diagnosis; parameters are computational constructs, not DSM-5-TR diagnostic labels.

3.1 Base Personality (OCEAN)

Parameter	Construct	Range	Transform	Sampling	Source
O	Openness	[0,1]	sigmoid	Joint (Vine Copula)	IPIP-FFM / HCP
C	Conscientiousness	[0,1]	sigmoid	Joint	IPIP-FFM / HCP
E	Extraversion	[0,1]	sigmoid	Joint	IPIP-FFM / HCP
A	Agreeableness	[0,1]	sigmoid	Joint	IPIP-FFM / HCP
N	Neuroticism	[0,1]	sigmoid	Joint	IPIP-FFM / HCP

Anti-double-counting rule: If Neuroticism is included as an OCEAN dimension, Negative Affectivity (from AMPD) must not be included as an independent parameter. The two are conceptually overlapping; including both would double-count the variance.

3.2 Dark Triad (Bifactor)

Parameter	Construct	Range	Transform	Sampling	Source
D	General Dark Core	[0,1]	sigmoid	Joint	SD3/DTDD bifactor
S_M	Machiavellianism specific	[0,1]	sigmoid	Joint	SD3/DTDD bifactor
S_P	Psychopathy specific	[0,1]	sigmoid	Joint	SD3/DTDD bifactor
S_N	Narcissism specific	[0,1]	sigmoid	Joint	SD3/DTDD bifactor

Modeling note: The general Dark Core factor (*D*) captures shared variance across all dark traits. Specific factors (*S_M*, *S_P*, *S_N*) capture trait-unique variance. The bifactor model is

estimated with orthogonal factors (specific factors uncorrelated with each other and with *D*), consistent with the meta-analytic SEM evidence .

3.3 Threat and Avoidance

Parameter	Construct	Range	Transform	Sampling	Source
θ_T	Threat sensitivity	[0,3]	log	Joint	Threat-learning literature
u_U	Uncertainty aversion	[0,3]	log	Joint	Anxiety/OCD literature

3.4 Reward Learning

Parameter	Construct	Range	Transform	Sampling	Source
α_R+	Positive learning rate	[0,1]	logit	Joint	RL task fitting
α_P-	Negative learning rate	[0,1]	logit	Joint	RL task fitting
β_choice	Inverse temperature	[0,10]	log-normal	Joint	Choice stochasticity

3.5 Intertemporal Choice

Parameter	Construct	Range	Transform	Sampling	Source
k_DD	Hyperbolic discount rate	>0	log-normal/gamma	Joint	Delay-discounting data

3.6 Dual-System Learning

Parameter	Construct	Range	Transform	Sampling	Source
w_MB	Model-based arbitration	[0,1]	logit	Joint	Kool two-step task
φ_per	Perseveration	[0,3]	log-normal	Joint	Repeat-choice data
η_vol	Volatility update	[0,1]	logit	Joint	Volatile learning tasks

3.7 Social Valuation

Parameter	Construct	Range	Transform	Sampling	Source
ω_social	Social value weight	[0,3]	log-normal	Joint	Trust/social reward data
λ_ineq	Inequity aversion	[0,3]	log-normal	Joint	UG/social choice
λ_norm	Moral norm sensitivity	[0,3]	log-normal	Joint	CNI/moral dilemmas
s_spite	Spite sensitivity	[0,3]	log-normal	Joint	Suspiciousness/social models

3.8 Perceptual Processing

Parameter	Construct	Range	Transform	Sampling	Source
τ_sens	Sensory precision	[0,3]	log-normal	Joint	Perceptual inference tasks
γ_sal	Salience gain	[0,3]	log-normal	Joint	Relevance/oddball tasks
κ_ctx	Context integration	[0,1]	logit	Joint	Contextual integration task

3.9 State and Inhibitory Control

Parameter	Construct	Range	Transform	Sampling	Source
m_t	Mood/state	[-3,3]	identity	AR(1)/state-space	Bipolar/depression state model
a_inhib	Inhibitor criterion/threshold	>0	log-normal	Joint	DDM/SSRT-compatible tasks

Calibration rule: No parameter may be assigned a value of min=0 or max=1 and interpreted as “disorder severity.” Parameters are computational constructs. Their recoverability must be assessed via parameter recovery before scale-up. Hierarchical Bayesian estimation does not guarantee parameter reliability (Katahira et al., 2024).

Part IV: Generative Engine — Vine Copula

4.1 Mathematical Implementation

Let $X = [X_1, X_2, \dots, X_d]$ be the d-dimensional vector of 21 psychological parameters (including the bifactor Dark Triad parameterization).

Step 1: Marginal Transformation. Convert empirical observations to uniform distributions via their empirical cumulative distribution functions (eCDFs): $U_i = F_i(X_i)$.

Step 2: Vine Structure Selection. Select the D-Vine or R-Vine tree structure that best fits the empirical dependence pattern using BIC or AIC. The vine decomposes the joint distribution into a cascade of bivariate pair-copulas .

Step 3: Pair-Copula Assignment. For each edge in the vine, select the bivariate copula family (Gaussian, Student-t, Clayton, Gumbel, Frank, Joe) that maximizes log-likelihood. Clayton copulas capture lower-tail dependence; Gumbel copulas capture upper-tail dependence; Joe copulas capture asymmetric upper-tail dependence.

Step 4: 32k Synthesis. Draw 32,000 samples from the fitted Vine Copula.

Step 5: Inversion. Apply the inverse CDFs to return the values to their true psychometric spaces: $\hat{X}_i = F_i^{-1}(\hat{U}_i)$.

Rigor Output: The result is a $32{,}000 \times 21$ matrix. Agent #14,052 might have clinically high general distress (DASS bifactor), severely impaired inhibitory control (SSRT index), and high general Dark Core (*D*) with specific narcissism elevation—traits that are probabilistically linked with correct tail behavior and asymmetric dependence.

4.2 Validation Against DT³ Findings (G2)

The Vine Copula output must be validated against the empirical DT³ findings:

Validation	**Metric**	**Pass Criterion**
Full 88-item network	Louvain community count	Dark Core Collapse emerges (all DT items merge into one community)
XGBoost predictors	Predictor ranking	Agreeableness → Machiavellianism; Empathy → Psychopathy; Age → Narcissism

Validation	Metric	Pass Criterion
SHAP interactions	Agreeableness × Empathy strength	Replicates approximately 0.1862
CKA representational similarity	Mean CKA vs. null	≈ 0.04 vs. ≈ 0.09
Semantic clustering	ARI	≈ 0.50 (Narcissism Fracture replicates)
Tail exceedance probabilities	Synthetic vs. empirical	Within 20% of empirical for extreme DASS-21 and SD3 scores

If the Vine Copula fails any of these validations, the copula family or vine structure must be revised before scale-up.

Part V: Mechanistic Cognitive Architecture

5.1 Module 1: Fehr-Schmidt Social Utility Function

An agent *i*'s perceived value of a systemic state is not merely its own resource accumulation (x_i), but is filtered through its social parameters:

[U_i(X) = x_i - \frac{\alpha_i}{N-1} \sum_{j \neq i} \max(x_j - x_i, 0) - \frac{\beta_i}{N-1} \sum_{j \neq i} \max(x_i - x_j, 0)]

- **α_i (Envy):** Bounds the agent's hostility toward richer neighbors. Correlated via Vine Copula to general Dark Core (*D*) and Neuroticism.
- **β_i (Guilt/Altruism):** Bounds the agent's willingness to share. Correlated to Agreeableness.

Identifiability Note: If recovery testing shows that α_i and β_i cannot be reliably recovered from the available Ultimatum/Trust trial counts, the utility function is simplified to a single-parameter inequity aversion index, and this simplification is flagged in every XAI trace as **[SIMPLIFIED MODEL]**.

5.2 Module 2: State-Space Dual-System Learning

When navigating the environment to maximize (U_i), the agent evaluates paths using a hybrid learning algorithm.

Model-Free (Habitual) System:

$$ Q_{MF}(s,a) \leftarrow Q_{MF}(s,a) + \alpha_{MF} [r + \gamma \max_{a'} Q_{MF}(s',a') - Q_{MF}(s,a)] $$

Model-Based (Goal-Directed) System:

$$ Q_{MB}(s,a) = \sum_{s'} P(s' \mid s,a) \max_{a'} Q_{MB}(s',a') $$

The Arbitration Node (*w_MB*):

$$ Q_{net}(s,a) = w_{MB,i} Q_{MB}(s,a) + (1 - w_{MB,i}) Q_{MF}(s,a) $$

Dynamic Decay Under Stress: $w_{MB,i}$ is not static. Under environmental stress—operationalized as activation of the DASS-21 general distress factor—$w_{MB,i}$ dynamically decays, suppressing Model-Based tree-search and forcing reversion to Model-Free habitual responses. The decay function is parameterized by the agent's general distress factor and validated against the Kool paradigm recovery data.

5.3 Module 3: Action Translation

Values translate into probabilities via a Softmax function:

$$ P(a_k) = \frac{\exp(\tau_i \cdot Q_{net}(s,a_k))}{\sum \exp(\tau_i \cdot Q_{net}(s,a_j))} $$

Where τ_i (Inverse Temperature) is derived from the agent's Openness to Experience and baseline exploration parameters.

5.4 Social Network Topology

Agents are linked using adjacency matrices to simulate social graphs (Small-World networks). This allows agents to play Trust Games and engage in resource trading simultaneously across the network graph. The adjacency matrix is a fixed structural parameter, not sampled from the Vine Copula.

Part VI: System Execution and Parallel Sandbox

6.1 JAX Tensor Vectorization Protocol

Object-Oriented Programming (OOP) with 32,000 separate class `Agent():` instances via standard Python loops will cause massive CPU bottlenecking. The environment is constructed using **Data-Oriented Design (DOD) via JAX**.

- All 32,000 Q-tables are flattened into a global tensor **Q** of shape `(32000, States, Actions)`.
- All state transitions and reward calculations are executed as parallel matrix multiplications on the GPU.

- **Dynamic *w_MB* decay** is implemented using `jax.vmap` mapped over the agent dimension, paired with `jax.lax.scan` to carry the temporal state. This avoids per-agent Python branching and compiles to an efficient XLA kernel.
- Each environment has its own market state and random key; `vmap` batches multiple environments for parallel execution .

G8 Benchmark: Before full implementation, write a localized 500-agent benchmark tracing 100 timesteps to measure device memory overhead and wall-clock execution limits. If the benchmark exceeds 60 seconds per 100 steps, the architecture must be revised before 32k scale-up.

6.2 Interrogation API: Deterministic Replay Architecture

To satisfy the XAI requirements without crashing device memory via host-to-device transfers, the Interrogation API will not run a continuous mirror. Instead, the simulation will utilize a **deterministic replay architecture**. Only the macro-state and RNG keys are logged at each tick. When a query is initiated at tick (t), the API re-simulates the target agent's specific trajectory from the nearest checkpoint on the CPU to extract (Q_{MB}), (Q_{MF}), and (\tau_i), guaranteeing zero latency overhead during the primary parallel GPU execution.

Part VII: XAI Protocol — Uncertainty-Aware Transparency

7.1 XAI Trace Schema

When a macro-phenomenon emerges in the simulation, the researcher can freeze the simulation at tick (t) and query any Agent (A_i). The XAI output traces the decision with uncertainty quantification:

1. **Log Output:** "Agent 14,052 betrayed Agent 8,991 at t=450."
2. **Utility Trace:** "Calculated (Q_{net}) for [Betray] was 8.4 [95% CI: 7.9–8.9], (Q_{net}) for [Cooperate] was 4.2 [95% CI: 3.6–4.8]."
3. **Algorithmic Breakdown:** "Despite (Q_{MB}) showing long-term value for cooperation, the agent's DASS-21 general distress factor spiked to 94th percentile due to recent resource loss. This forced arbitration weight (w_{MB,i} \to 0.05) [recovery ρ = 0.71, posterior shrinkage factor = 0.68], blinding the agent to the future and relying purely on immediate Model-Free ((Q_{MF})) greed."
4. **Social Trace:** "Furthermore, baseline Fehr-Schmidt parameter α_i = 0.82 [95% CI: 0.61–1.00, identifiability status: marginal] generated negative utility for cooperating with an agent perceived to have higher resources."

7.2 Uncertainty Flags

Flag	Meaning
[SYNTHETIC PRIOR]	Parameter not directly estimated from human data; prior derived from literature
[LOW RECOVERABILITY]	Parameter recovery ρ < 0.7
[HIGH SHRINKAGE]	Posterior shrinkage factor < 0.7
[SIMPLIFIED MODEL]	Parameter simplified due to identifiability constraints

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Part VIII: Master Execution Roadmap — Sprint 0 Gated

8.1 Sprint 0: Verification & Recovery (Mandatory Gate)

All progression to Sprint 3 is halted until G1–G9 are passed.

Gate	Action	Owner	Deliverable	Pass/Fail Criterion
G1	Parameter recovery for ALL 21 parameters	Kian	Bias/RMSE/recovery report	≥70% of parameters with rank-order recovery ρ > 0.7
G2	Vine Copula fidelity test against empirical N≈500 matrix	Kian	Wasserstein/energy distance + tail metrics	Tail exceedance probabilities within 20% of empirical
G3	SD3/DTDD bifactor CFA on actual data	Sabihe	CFA output (3-factor vs. bifactor)	Adopt empirically superior structure
G4	DASS-21 bifactor CFA on actual data	Sabihe	CFA output	Pre-register measurement model (bifactor preferred)
G5	Kool two-step *w_MB* recovery	Kian	Recovery comparison	*w_MB* recoverable (ρ

Gate	Action	Owner	Deliverable	Pass/Fail Criterion
				> 0.6) in Kool paradigm
G6	Fehr-Schmidt α/β recovery	Kian	Recovery report	α/β rank-order recovery ρ > 0.5; else simplify model
G7	Stop-Signal RT availability check	Kian	Data audit	Document whether DDM is recoverable; fallback to SSRT
G8	JAX 500-agent benchmark (100 steps)	Kian	Wall-clock measurement	< 60s per 100 steps on target hardware
G9	Data provenance matrix	Both	Parameter → dataset → license mapping	No parameter without a documented source

8.2 Sprint 1: Empirical Data Ingestion & HBA Fitting

Sabihe Khorshidi: Clean and standardize the HCP OCEAN, SD3/DTDD bifactor, and DASS-21 bifactor datasets. Identify extreme outliers. Map the conceptual linkages between psychological traits and algorithmic hyper-parameters.

Kian Mansouri Jamshidi: Write parsers for the BIDS/TSV formats downloaded from OpenNeuro (ds000030, ds003745) and OSF. Implement Bayesian fitting scripts (PyMC or Stan) to extract individual parameters from the raw human trial logs.

8.3 Sprint 2: The Vine Copula Core

Sabihe Khorshidi: Review the aggregate parameter distributions. Validate that the correlations and tail dependencies make psychological sense.

Kian Mansouri Jamshidi: Build the Vine Copula pipeline. Feed the validated empirical parameters in, and generate the master `32k_Agent_Genome.h5` matrix. Validate against DT³ findings (G2).

8.4 Sprint 3: The Engine Sandbox (Gated by Sprint 0)

Kian Mansouri Jamshidi: Engineer the JAX vectorized tensor environment. Build the state-space matrices, the reward distribution algorithms, and the network graph adjacency arrays.

Sabihe Khorshidi: Design the “Scenarios” (e.g., Resource Scarcity, Wealth Inequality shocks). Define how environmental state changes map to the agents’ DASS-21 general distress triggers.

8.5 Sprint 4: The XAI Dashboard & Testing

Kian Mansouri Jamshidi: Build the Interrogation API with deterministic replay. Ensure that at every tick, matrices of (Q), (\tau), and (w_{MB}) can be logged and visualized with uncertainty flags.

Both Leads: Run the first 32,000-agent systemic test. Monitor the emergent dynamics, pause the simulation, and use the XAI framework to explain macro-behaviors using the deterministic micro-equations.

Part IX: Data Directory and Provenance

9.1 Current Data Assets

Directory	Content	Parameter Coverage
`XAI_Markov_TwoStep_Data`	Two-step task behavioral data	*w_MB*, α_R+, α_P-, β_choice
`3T_TFMRI_inscanner_performance_data`	HCP in-scanner task performance	Covariates
`HCP_YA_subjects_2026_10_06_10_54_40.csv`	HCP subject metadata	Demographics
`IPIP-FFM-data-8Nov2018`	Big Five personality data	O, C, E, A, N
`participants.tsv`	BIDS participants file	Demographics
`SD3`	Short Dark Triad data	*D*, *S_M*, *S_P*, *S_N*
`XAI_Behavioral_Data`	General behavioral data	Task-specific

Directory	Content	Parameter Coverage
`XAI_Delay_Discounting_Data`	Delay discounting task	*k_DD*
`XAI_Mental_Health_DASS`	DASS-21 scores	General distress, m_t
`XAI_Risk_Impulsivity_Data`	BART/Stop-Signal	*a_inhib* (SSRT or DDM)
`XAI_Social_Games_Data`	Trust/Ultimatum games	α_i, β_i, λ_ineq, ω_social

9.2 Data Provenance Requirements

Every parameter must have a documented source. Parameters lacking direct empirical grounding are designated **[SYNTHETIC PRIOR]** and flagged in the XAI trace. The G9 provenance matrix must map:

- Parameter → Dataset → License/DUA → Access date → Preprocessing code
- Parameters requiring restricted data (e.g., HCP kinship) must be flagged if restricted access is not secured.

Part X: Non-Negotiable Scientific Constraints

1. **Determinism language:** The architecture is described as "within-agent deterministic, between-agent stochastic, parameterized by a fixed random seed." The word "deterministic" is never used to describe the population as a whole.
2. **MBTI removal:** MBTI is entirely removed from all specifications and downstream artifacts.
3. **100% transparency removal:** The phrase "100% transparent glass-box" is replaced with "uncertainty-aware mechanistic transparency."
4. **Dark Triad bifactor:** The Dark Triad is modeled as a bifactor structure (general Dark Core + specific factors), not as three independent Copula dimensions.
5. **Gaussian Copula removal:** The Gaussian Copula is replaced with a Vine Copula (D-Vine or R-Vine) that can represent tail dependence and asymmetric interactions.
6. **Scale-up gating:** No 32k simulation is run until G1–G9 are passed. A 500-agent pilot must reproduce the DT³ findings (Dark Core Collapse, Narcissism Fracture, agreeableness/empathy two-branch structure) before 64× scale-up.
7. **Parameter recovery:** Every parameter in the Copula must have rank-order recovery $\rho > 0.7$ (or documented exemption with justification). Parameters failing recovery are either simplified, flagged as synthetic priors, or removed.
8. **DASS-21 bifactor:** Dynamic modulation of *w_MB* uses the general distress factor from the bifactor model, not three orthogonal factors.

9. **Stop-Signal fallback:** If trial-level RTs are unavailable, DDM parameters are not fitted; SSRT is used as the inhibitory control index.
10. **Kool paradigm for *w_MB*:** The Kool two-step paradigm is the primary source of *w_MB* values, with the Daw paradigm retained for comparison and validation.
11. **No arbitrary synthetic agents:** All parameters are tethered to empirical data or documented synthetic priors. No parameter is sampled from an uninformed distribution.
12. **Reproducibility:** Every agent has a unique ID and seed. Trial-level data is stored. Generator version and seed list are immutable. Code is archived with commit hash or release tag.

Part XI: Deliverables

ID	Deliverable	Status	Content
D1	Psychological Framework v1.2	Completed	Full 21-parameter dictionary, bifactor CTAs
D2	DSM-5-TR/HiTOP Clinical Anchor Map	Completed	Dimensional anchors, not binary diagnoses
D3	Mechanism Matrix	Completed	Construct → parameter → behavior
D4	Parameter Dictionary	Completed	Range/transform/distribution/evidence
D5	Task Battery Specification	Completed	Task → construct → outputs → protocol
D6	Vine Copula Generator Specification	Completed	D-Vine/R-Vine, pair-copula selection, validation
D7	Comorbidity/Interaction Rules	Completed	Dimensional sampling, no independent Bernoulli
D8	Hypothesis Register	Completed	H1–H10 + falsification criteria

ID	Deliverable	Status	Content
D9	Validation Protocol	Completed	Recovery/identifiability/human benchmark
D10	Human Data Acquisition Matrix	Completed	HCP/ABCD/OpenNeuro/NDA + limitations
D11	Trial-Level Trace Schema	Completed	State, belief, uncertainty, value, PE, utility, policy, action, update
D12	Preregistration Checklist	Completed	Freeze/QC/reproducibility
D13	Raw Human Re-analysis	**Pending**	Download/harmonize/model open datasets (Gates G1–G9)
D14	Full 32k Main Simulation	**Pending**	Run technical pipeline (post-freeze by Kian)

Part XII: References

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Final Statement: This document is the single, self-contained master specification for Project 32k-XAI. It supersedes all prior specifications and revisions. No progression to Sprint 3 (32k Sandbox) is permitted until Gates G1–G9 are documented and passed. The 32k simulation is a scale-up of validated components, not a validation step. Scaling unvalidated components would produce a 32,000-agent monument to measurement error. The revised specification resolves all contradictions with the published DT³ article and establishes the scientifically rigorous foundation required for DARPA-grade simulation.